Supervised Learning Report Rubric
CS7641: Machine Learning
Fall 2026

Schedule

- **9/28/2026 (Monday):** SL Grading Begins & Overview of Rubric
- **9/30/2026 (Wednesday):** SL First 2 Calibration
- **10/2/2026 (Friday):** SL Calibration with Examples and Head TAs Check
- **10/6/2026 (Tuesday):** SL 33–50% Completed
- **10/9/2026 (Friday):** SL 50–80% Completed
- **10/13/2026 (Tuesday):** SL 100% Completed
- **10/14/2026 (Wednesday):** SL Posted

Total: 100 points + 5 extra credit

This rubric is aligned to the Fall 2026 Supervised Learning assignment specification.

Each scored line item has a fixed point value and is graded **all-or-nothing**. Award the listed points only when the required evidence is explicit, correct, and supported by the submitted work.

The main report must stand alone as a formal technical paper. The reproducibility sheet may contain extended EDA, implementation details, environment information, and supporting materials, but it does not replace substantive analysis in the main report.

Required Task

Students must complete the following task:

- **Dataset:** Diabetes Health Indicators Dataset derived from the 2015 Behavioral Risk Factor Surveillance System (BRFSS).
- **Required file:** `diabetes_012_health_indicators_BRFSS2015.csv`
- **Do not accept:** `diabetes_binary_health_indicators_BRFSS2015.csv` or `diabetes_binary_5050split_health_indicators_BRFSS2015.csv`.
- **Task:** Multiclass classification.
- **Target:** `GenHlth`, representing self-reported General Health: 1 = Excellent, 2 = Very Good, 3 = Good, 4 = Fair, 5 = Poor.
- **Full dataset size:** 253,680 instances.
- **Working sample:** Approximately 20,000 instances.
- **Required metrics:** Accuracy, Macro-F1, balanced accuracy, confusion matrix, and per-class precision, recall, and F1.

The `GenHlth` column must not appear among the predictor variables.

Sampling Expectation

Students must create and justify an approximately **20,000-instance** working dataset.

A stratified sample is the default expectation because `GenHlth` is imbalanced. A simple random sample is acceptable only if the student demonstrates that all five classes are preserved and that the sample reasonably reflects the full-data class distribution.

Students should not use the first 20,000 rows without compelling justification.

The report or reproducibility materials should identify:

- full and sampled dataset sizes;
- full and sampled class distributions;
- sampling method and rationale;
- implications for less-represented classes; and
- exact random seed and sampling procedure.

Minimum Hyperparameter Exploration Standard

When a rubric item requires multiple settings, the default expectation is at least **three distinct and reasonable values** for the relevant scalar hyperparameter unless a clear computational justification is provided.

The tested values should span enough range to reveal meaningful changes in underfitting, overfitting, or performance saturation.

For example, $k = 3, 15, 75$ may demonstrate meaningfully different neighborhood sizes, whereas $k = 5, 6, 7$ generally does not without prior evidence that such a narrow region warrants detailed exploration.

Computationally expensive models, especially SVMs, may use a smaller search when runtime constraints are documented and the remaining comparison is still sufficient to justify the final model.

Model Expectations

Students must evaluate:

- Decision Trees;
- k-Nearest Neighbors;
- Support Vector Machines; and
- Neural Networks using both scikit-learn and PyTorch.

TAs should not expect one universal model ranking. Grade whether students explain observed behavior using evidence from metrics, learning curves, model-complexity curves, runtime, class imbalance, and error patterns.

BRFSS contains binary, ordered, and differently scaled health-related variables. Scaling therefore matters for distance- and margin-based methods, while Decision Trees are comparatively insensitive to monotonic scaling. Neural Networks may be particularly sensitive to preprocessing and SGD configuration.

Decision Trees

Students must use a complexity-control mechanism such as `max_depth`, `min_samples_leaf`, `min_samples_split`, or `ccp_alpha`.

Expect a model-complexity curve and an explanation of the selected tree in terms of underfitting, overfitting, bias, or variance.

k-Nearest Neighbors

Students must use leakage-safe scaling, state the distance metric and weighting approach, and explore meaningfully different values of k.

Students should discuss both predictive behavior and prediction-time cost.

Support Vector Machines

Students must evaluate at least two kernels, such as linear and RBF, use leakage-safe scaling, tune C, and tune γ where appropriate.

Reduced SVM searches are acceptable when computational constraints are documented and enough evidence remains to support the final choice.

Neural Networks

Two implementations are required:

- **scikit-learn:** `MLPClassifier`
- **PyTorch:** compact MLP

Both must use **plain SGD only**.

For scikit-learn:

```
solver='sgd'
momentum=0
nesterovs_momentum=False
```

For PyTorch, use `torch.optim.SGD` without momentum.

Adam, AdamW, Adagrad, RMSprop, momentum, Nesterov momentum, and other adaptive optimizers are not allowed.

Students must document architecture, learning rate, batch size, epochs, regularization, stopping behavior, and relevant seeds.

Epoch-based training and validation curves are required for both implementations.

Interpreting Model Performance

Do not grade students based on reaching a particular predictive score.

For this assignment, model performance in the approximate **40–60% range** for Accuracy, Macro-F1, balanced accuracy, or related evaluation measures may be entirely reasonable depending on the model, metric, class imbalance, preprocessing, and experimental configuration.

A score in this range should **not automatically be interpreted as evidence that the implementation is incorrect or that the model is "bad."**

The important question is whether the student can explain what the result means.

For example, a model with approximately 50% performance may still support a strong analysis if the student demonstrates that:

- the model substantially exceeds an appropriate baseline;
- performance differs meaningfully across Accuracy, Macro-F1, and balanced accuracy;
- majority classes are predicted reasonably while less-represented classes remain difficult;
- the learning curve suggests high bias or limited benefit from additional data;

- the complexity curve shows that additional model capacity does not materially improve validation performance;
- stronger regularization improves generalization while reducing training performance;
- class overlap or feature limitations appear to constrain separability;
- the model has a favorable runtime or interpretability trade-off despite lower predictive performance; or
- the result exposes a meaningful limitation of the algorithm on this dataset.

A student should therefore avoid statements such as:

> "The Decision Tree only achieved 48% accuracy, so Decision Trees are bad for this problem."

A stronger interpretation would explain **why** performance is limited and what the limitation implies. For example:

> "The Decision Tree reached approximately 48% accuracy and substantially lower Macro-F1. The class-level results show that most errors occur in the less-represented Fair and Poor categories. Increasing tree depth improves training accuracy without comparable validation improvement, suggesting variance rather than insufficient model capacity. The result implies that additional tree complexity alone is unlikely to resolve the observed class-level errors."

Similarly, a model with a modest aggregate score may reveal useful information about the dataset. Poor predictive performance can be an important experimental result when the student uses curves, class-level metrics, baselines, and algorithmic theory to explain **why the model struggles and what follows from that result**.

TAs should reward this type of analysis.

Conversely, a student should **not** receive strong analytical credit merely for obtaining a high score if the report does not explain why the model performs well, whether performance is consistent across classes, or whether the result is supported by leakage-safe evaluation.

Always interpret model scores relative to:

- baseline performance;
- the metric being reported;
- class distribution;
- other required algorithms;
- learning and complexity curves;
- runtime;
- class-level error patterns; and
- the student's original hypothesis.

TA Start Check

Complete this short check when beginning each submission.

☐ Both required PDFs are present and readable.

☐ Correct dataset and required file `diabetes_012_health_indicators_BRFSS2015.csv`are used.

☐ `GenHlth` is treated as a five-class classification target and is excluded from the predictors.

☐ Approximately 20,000 instances are used for the working dataset.

☐ Overleaf **Edit Link** is present and accessible.

☐ GT Enterprise GitHub repository and final commit hash are accessible.

☐ Reproducibility instructions and sampling information are present.

☐ Required AI Use Statement is present.

☐ Report is within the 8-page scoring limit.

☐ Report reads as a technical paper rather than a checklist or bullet-only draft.

Submission Verification Rule

If a required submission artifact or required access is missing or unverifiable:

1. Continue grading the available work.
2. Provide normal substantive feedback.
3. Record the rubric score the work would otherwise earn.
4. Enter a temporary overall score of **0**.
5. Identify the submission/access issue that must be corrected.

Once the student provides and verifies the required material, replace the temporary zero with the **rubric score already earned**.

Do not apply an additional point deduction for the same access or verification issue.

Using the wrong dataset, wrong required file, or wrong target is a substantive failure to complete the assigned task rather than a correctable access issue.

Overleaf Edit Access

The Edit Link allows the instructional team to inspect project/version history **if a provenance, authorship, version-history, or AI-use concern arises**.

TAs do not need to perform a detailed Overleaf-history review for every submission.

Integrity and Provenance Triage

Turnitin scores, AI indicators, writing-style changes, large Overleaf paste events, unusual Git history, AI disclosures, and agent-generated repository files are **signals for review, not automatic findings of misconduct**.

Additional provenance review is appropriate when a specific concern exists. Examples include:

- substantial or unusual Turnitin overlap in analysis or conclusions;
- AI Use Statement that appears inconsistent with the submitted work;
- large unexplained blocks of report content appearing at once;

- repository files containing generated analysis, conclusions, prompts, or copied course materials;
- suspicious similarity with another student's analysis, figures, or wording; or
- other concrete evidence that authorship or provenance should be checked.

What Obvious AI Use May Look Like

TAs should be especially attentive to **internal inconsistencies or artifacts that suggest generated material was inserted without being verified**.

Stronger signals worth checking include:

- **Prompt or assistant residue.** Examples include phrases such as "Here is your revised analysis," "As an AI," "you should include," "insert figure here," or other instructions that appear to have been copied directly from an AI interaction.
- **Meta-commentary written to the student rather than report prose.** For example, text that tells the author what to do instead of presenting the analysis itself.
- **Claims about experiments that do not exist.** The prose may discuss a model, hyperparameter sweep, statistical test, figure, or result that is not present in the submitted code, figures, tables, or reproducibility materials.
- **Numbers that conflict with the student's own figures or tables.** For example, the discussion states that SVM obtained 72% accuracy when the corresponding table reports 54%.
- **Analysis that is generic enough to fit any dataset.** Examples include multiple paragraphs explaining textbook definitions of bias and variance without connecting them to the student's actual learning curves, metrics, or BRFSS results.
- **Highly specific interpretations unsupported by the experiment.** For example, asserting that a feature caused a particular error pattern when no feature analysis supporting that statement was performed.
- **Fabricated or unverifiable citations.** Examples include references with nonexistent titles, incorrect authors, impossible publication information, or citations that do not support the claim attributed to them.
- **Citation style that changes abruptly or references that appear only in generated prose.**
- **Terminology inconsistent with the assignment.** Examples include repeatedly referring to binary classification, ROC-AUC as the primary required metric, a different target variable, a different dataset, or algorithms not actually used.
- **Code/report mismatch.** For example, the report describes RBF SVM tuning over C and γ, while the repository contains only a single linear SVM run.
- **Figure/report mismatch.** For example, prose claims overfitting while the presented learning curve does not show a meaningful training-validation gap.
- **Generated repository artifacts that contain report-ready analysis.** An agent-generated summary, prompt transcript, context file, or planning artifact that contains prose nearly identical to the submitted analysis may warrant closer review.
- **Large blocks of polished analysis appearing in Overleaf with little or no intermediate development**, particularly when combined with other inconsistencies.

These signals are more meaningful when **multiple pieces of evidence agree**.

The following by themselves are **weak signals and should not be treated as evidence of prohibited AI use**:

- polished grammar;
- formal vocabulary;
- sentence structure;

- use of particular punctuation;
- a sudden improvement in writing quality;
- a high AI-detector score;
- a single large paste into Overleaf;
- use of common technical phrases; or
- disclosure that an AI tool was used.

The strongest concern is generally not that prose "sounds like AI." The stronger concern is when the submitted analysis demonstrates **a mismatch between what is written and the experiments the student actually performed**.

When Additional Review Is Needed

If additional review is needed, inspect relevant Overleaf history, repository history, commit information, and disclosed AI use.

Agent artifacts such as `CONTEXT_SUMMARY.md`, `CLAUDE.md`, prompt logs, generated checklists, or tool-context files are not violations by themselves. Record and escalate them only when their contents raise a specific concern.

Likewise, a student who discloses using AI to create a checklist from assignment materials should **not receive an automatic grading penalty**. Record the disclosure and escalate it if the described use may conflict with course AI or course-material policies.

When a meaningful concern remains:

1. finish normal academic grading whenever possible;
2. document the concern factually;
3. preserve the relevant report page, Turnitin match, repository path/commit, Overleaf evidence, or AI disclosure;
4. send the evidence to the Head TA/instructor; and
5. do not assign an ad hoc misconduct penalty or accuse the student in ordinary grading feedback.

Tracking Only

Record the following for instructor review only:

- **Blue/yellow figures:** Do graphs and figures use the requested high-contrast blue/yellow scheme? Yes / No
- **Flummoxed:** Does the report contain the word "flummoxed" multiple times? Yes / No

These items do **not** affect the student's score.

Rubric Points

All criteria below are scored all-or-nothing.

Compliance (16 points)

Data Compliance (3 points)

- **(2 pts) Sample size.**
  Award if: The report uses an approximately 20,000-instance working dataset and clearly documents its size. For example, a stratified sample of 19,800 or 20,200 instances is reasonable if the choice is clear and reproducible.
  Do not award if: The student runs the required experiments on the full 253,680-instance dataset, uses an arbitrarily small subset such as 2,000 instances, or uses a substantially different sample size without explanation.

- **(1 pt) Source and target.**
  Award if: The required BRFSS file is used and `GenHlth` values 1–5 are consistently treated as the multiclass target.
  Do not award if: The student uses one of the binary diabetes files, predicts a different target such as `Diabetes_012`, collapses `GenHlth` into a binary label, or includes `GenHlth` among the predictors.

Submission Access (4 points)

- **(2 pts) Enterprise Overleaf.**
  Award if: A working Edit Link is provided for the student's Georgia Tech / Enterprise Overleaf project and the project can be accessed by course staff.
  Do not award if: The link is missing, is only a View Link, is inaccessible, or points only to a personal/non-Enterprise project. Apply the temporary-zero verification rule rather than treating the access issue as only a 2-point loss.

- **(2 pts) Enterprise GitHub.**
  Award if: A valid final commit hash from the student's GT Enterprise GitHub repository is provided and the corresponding repository/commit is accessible.
  Do not award if: The hash is missing or invalid, the repository cannot be accessed, or only a personal GitHub repository is supplied. Apply the temporary-zero verification rule.

Reproducibility (3 points)

- **(2 pts) Run instructions.**
  Award if: Commands, environment requirements, data paths, and seeds are sufficient for a TA to understand how the experiments would be reproduced on a standard Linux machine. For example, the student provides environment/package instructions plus the command or script order needed to regenerate outputs.
  Do not award if: The student provides only vague statements such as "run the notebook," omits required dependencies or data paths, or provides no usable execution order.

- **(1 pt) Sampling procedure.**
  Award if: The student provides the sampling method, sample size, random seed, and enough detail or code to recreate the sample.
  Do not award if: The report says only "we randomly sampled 20,000 rows," gives no seed/procedure, or otherwise makes the sampled dataset impossible to reproduce.

Figure Legibility (2 points)

- **(2 pts) Readable figures and tables.**
  Award if: Figures, tables, axis labels, legends, captions, and text are readable at normal grading zoom. For example, learning-curve legends and tick labels can be read without zooming to 200–300%.
  Do not award if: Multiple visual elements are too small, blurry, clipped, missing labels, or otherwise difficult to evaluate reliably.

Attribution (4 points)

- **(2 pts) Outside sources.**
  Award if: At least two relevant peer-reviewed sources beyond course materials are used to justify methodology or interpretation. For example, a peer-reviewed source may support metric selection for imbalanced classification or a methodological decision involving SVMs, pruning, or leakage.
  Do not award if: The student cites only documentation, blogs, textbooks, lecture slides, or arXiv/preprint material; provides fewer than two qualifying peer-reviewed sources; or lists papers in References without actually using them in the analysis.

- **(2 pts) AI usage statement.**
  Award if: The report identifies AI tools used, describes what they assisted with, and affirms that the student reviewed and understood assisted content.
  Do not award if: The AI statement is absent, says only "AI was used," fails to identify tools or use cases, or does not confirm review/understanding of assisted content.

Study Design (14 points)

EDA Evidence (4 points)

- **(2 pts) Full-data EDA.**
  Award if: The report includes dataset shape, relevant feature characteristics, missingness or absence of missingness, class counts/proportions, and at least one observation that informs later modeling. For example, the student may note strong class imbalance or substantial differences in feature scales and connect those observations to metric or preprocessing choices.
  Do not award if: EDA consists only of generic summary statistics, omits the `GenHlth` class distribution, or presents plots without explaining how they affect the experimental design.

- **(2 pts) Sample EDA.**
  Award if: Class counts and proportions for the approximately 20,000-instance sample are reported and compared with the full-data distribution.
  Do not award if: The sample distribution is missing, only percentages or counts are shown without sufficient context, or the student never checks whether the sampling strategy preserves the five classes.

Sampling Design (4 points)

- **(3 pts) Sampling rationale.**
  Award if: Stratification is used and justified, or an alternative procedure is defended with evidence showing that all classes remain appropriately represented. The student discusses consequences for minority classes.
  Do not award if: The student simply states that 20,000 rows were sampled at random, does not compare class distributions, or uses a non-stratified sample without demonstrating that minority-class representation remains reasonable.

- **(1 pt) Ordering effects.**
  Award if: Sampling is randomized or stratified using a documented seed, or another procedure is convincingly justified.
  Do not award if: The student uses `head(20000)`, the first 20,000 rows, or another order-dependent selection without evidence that ordering is harmless.

Metrics and Hypothesis (4 points)

- **(2 pts) Metric justification.**
  Award if: The report explains why Accuracy, Macro-F1, balanced accuracy, confusion matrices, and class-level precision/recall/F1 are useful for this imbalanced five-class problem. For example, the student explains that overall Accuracy may remain high while performance on Fair or Poor health classes is weak.
  Do not award if: Metrics are merely listed, Accuracy is treated as sufficient by itself, or Macro-F1/balanced accuracy are reported without explaining what they reveal.

- **(1 pt) Early hypothesis.**
  Award if: A clear, testable hypothesis appears after EDA and before tuned results and connects dataset

properties to supervised-learning theory. For example, "RBF SVM should outperform linear SVM because the EDA suggests nonlinear class structure, although it will likely require greater runtime."
Do not award if: The hypothesis is a generic goal such as "SVM will do well," is introduced after the model results, or contains no connection to dataset properties or course concepts.

- **(1 pt) Hypothesis revisit.**
Award if: The discussion or conclusion explicitly determines whether the hypothesis was supported, contradicted, or partially supported using experimental evidence.
Do not award if: The report states an early hypothesis but never returns to it, or merely says "the hypothesis was correct" without supporting evidence.

Validation (2 points)

- **(1 pt) Test holdout.**
Award if: A single held-out test set is reserved for final evaluation after model/hyperparameter selection.
Do not award if: Test-set results are repeatedly used to choose k, tree depth, SVM parameters, NN architecture, or other model settings.

- **(1 pt) Leakage-safe validation.**
Award if: Tuning occurs using training-only validation, and learned preprocessing such as scaling is fit only on training data or training folds.
Do not award if: Scaling is applied to the entire dataset before splitting, `GenHlth` leaks into the features, or test observations influence preprocessing/model selection.

Decision Trees (14 points)

- **(2 pts) Complexity control.**
Award if: A pruning or regularization mechanism is identified and its effect on model capacity is explained. Examples include `max_depth`, `min_samples_leaf`, or `ccp_alpha`.
Do not award if: The student uses an unconstrained tree without discussing complexity control, or merely reports final depth after training without explaining how complexity was managed.

- **(3 pts) Learning curve.**
Award if: Training and validation performance are shown across at least four training sizes and interpreted using bias/variance reasoning. For example, a persistent training-validation gap may be identified as evidence of variance.
Do not award if: The curve has too few training sizes without justification, shows only validation performance, or is described only as "performance increases with more data" without interpretation.

- **(3 pts) Complexity curve.**
Award if: At least three meaningful values of a tree-complexity parameter are compared, and the curve supports the final model choice. For example, depth may be varied broadly enough to show underfitting at shallow depths and overfitting at larger depths.
Do not award if: Only one or two values are tested without justification, the values are too narrow to show a meaningful capacity change, or the selected tree is unrelated to the complexity evidence.

- **(2 pts) Final evaluation.**
Award if: Accuracy, Macro-F1, balanced accuracy, and meaningful class-level evaluation are reported. A modest score is acceptable when the student correctly analyzes what limits performance and what the result implies.
Do not award if: Required aggregate metrics are missing, the student provides a confusion matrix but does not discuss any class-level behavior, or low performance is dismissed simply as evidence that the algorithm is "bad."

- **(2 pts) Runtime.**
Award if: Fit and prediction time are reported and at least one relevant trade-off is discussed, such as interpretability or low prediction cost versus predictive performance.
Do not award if: Only a fit time is listed with no context, or runtime appears in a table but is never discussed.

- **(2 pts) Tuning quality.**
  Award if: The final tree is supported by meaningful validation or complexity evidence.
  Do not award if: A final depth or pruning value appears arbitrary, is chosen from test performance, or is selected without sufficient exploration.

kNN (14 points)

- **(2 pts) Scaling and distance.**
  Award if: Scaling is leakage-safe, and the distance metric and weighting strategy are stated and justified. For example, the student explains why standardization is needed before Euclidean distance because BMI and binary indicators exist on different numeric scales.
  Do not award if: No scaling is used or discussed, scaling occurs before the split, or the distance/weighting choices are left at defaults without being identified.

- **(2 pts) k exploration.**
  Award if: At least three meaningfully different neighborhood sizes are tested. For example, small, moderate, and relatively large k values are used to expose changes in locality and smoothing.
  Do not award if: Only one k is tested, or values such as $5, 6, 7$ are used without a reason that such a narrow region is scientifically meaningful.

- **(2 pts) Learning curve.**
  Award if: Training and validation performance are shown across at least four training sizes and interpreted.
  Do not award if: The learning curve is missing, too sparse without justification, or shown without discussing whether additional data appears useful.

- **(2 pts) Complexity curve.**
  Award if: Validation performance across k values supports the final choice and is interpreted in terms of model complexity/locality.
  Do not award if: The curve is shown but the selected k is not connected to it, or there is no discussion of how neighborhood size affects behavior.

- **(3 pts) Final evaluation.**
  Award if: Accuracy, Macro-F1, balanced accuracy, and meaningful class-level evaluation are reported. Modest performance receives full credit when the analysis correctly explains locality, imbalance, class overlap, or other evidence-supported limitations.
  Do not award if: Required metrics are missing, minority-class behavior is ignored despite poor per-class recall/F1, or the student simply labels kNN ineffective based on a modest aggregate score.

- **(2 pts) Runtime.**
  Award if: Fit and prediction times are reported and the characteristic prediction-time cost of kNN is discussed.
  Do not award if: The student calls kNN "fast" solely because fit time is small or omits prediction-time discussion.

- **(1 pt) Sensitivity discussion.**
  Award if: At least one kNN-specific issue is discussed, such as scale, locality, irrelevant features, mixed binary/numeric features, neighborhood density, or class imbalance.
  Do not award if: The analysis contains no kNN-specific interpretation and could apply unchanged to any model.

SVM (14 points)

- **(2 pts) Kernel comparison.**
  Award if: At least two kernels are actually evaluated and meaningfully compared. For example, linear and RBF results are contrasted in performance, runtime, or class-level behavior.
  Do not award if: Only one kernel is used, or a second kernel is mentioned in methodology without presenting comparable results.

- **(2 pts) Scaling.**
  Award if: Leakage-safe feature scaling is applied consistently and described.

Do not award if: Scaling is missing, fit using the full dataset before splitting, or applied inconsistently between kernels.

- **(2 pts) Learning curve.**
  Award if: Training and validation performance are shown across at least four training sizes and interpreted.
  Do not award if: The curve is missing, contains too few training sizes without justification, or is presented without explaining model behavior.

- **(2 pts) Complexity curve.**
  Award if: C and/or γ are varied across meaningful settings and the result is interpreted. A smaller sweep is acceptable when supported by runtime evidence.
  Do not award if: A single C or γ is used, the explored values are too narrow without justification, or a reduced search is blamed on runtime without reporting evidence.

- **(3 pts) Final evaluation.**
  Award if: Accuracy, Macro-F1, balanced accuracy, and meaningful class-level evaluation are reported. The quality of the analysis does not depend on the SVM reaching a predetermined performance threshold.
  Do not award if: Required metrics are missing, the model is declared best or worst based only on Accuracy, or modest performance is reported without investigating kernel behavior, class-level errors, or tuning evidence.

- **(1 pt) Runtime.**
  Award if: Fit/prediction time and relevant kernel-cost differences are discussed.
  Do not award if: Runtime is absent, or the report states that RBF was expensive without providing supporting runtime information.

- **(2 pts) Tuning quality.**
  Award if: Kernel choice, C, and relevant nonlinear parameters are explored sufficiently to support the final configuration.
  Do not award if: Kernel or parameter choices appear arbitrary, γ is never explored for RBF without justification, or the held-out test set drives tuning.

Neural Networks (14 points)

If only one required implementation is completed, award at most half of this section.

- **(3 pts) SGD setup.**
  Award if: Both implementations use plain SGD and document architecture, learning rate, batch size, epochs, stopping behavior, and regularization. For scikit-learn, `solver='sgd'`, `momentum=0`, and `nesterovs_momentum=False` are used; PyTorch uses plain `torch.optim.SGD` without momentum.
  Do not award if: Adam, AdamW, RMSprop, Adagrad, momentum, Nesterov, or another disallowed optimization method is used; or the optimizer/training configuration cannot be verified.

- **(2 pts) Epoch curves.**
  Award if: Training and validation loss or another justified metric are shown over epochs for both implementations and used to discuss convergence or stability.
  Do not award if: Epoch curves are shown for only one implementation, only training loss is shown with no validation information, or the curves are presented without interpretation.

- **(2 pts) Learning curve.**
  Award if: Training and validation performance are shown across at least four training sizes and interpreted.
  Do not award if: The learning curve is missing, too sparse without justification, or does not support any reasoning about data sufficiency or generalization.

- **(2 pts) Complexity curve.**
  Award if: At least three meaningful NN settings are compared using a capacity, learning-rate, or regularization dimension and support the final configuration.
  Do not award if: Only one architecture is used, multiple parameters are changed simultaneously so that the comparison is uninterpretable, or the explored settings do not support the selected model.

- **(2 pts) Final evaluation.**
  Award if: Accuracy, Macro-F1, balanced accuracy, and meaningful class-level evaluation are reported. A neural network in the 40–60% range can still receive full credit when the student uses epoch curves, class-level

behavior, and optimization evidence to explain the result under the SGD-only constraint.
Do not award if: Only final Accuracy is reported, class-level behavior is not examined, or low performance is attributed to "neural networks being bad" without examining convergence, learning rate, architecture, regularization, or optimization behavior.

- **(2 pts) Implementation comparison.**
Award if: scikit-learn and PyTorch are compared in terms of optimization behavior, stability, runtime, tunability, and/or final performance. For example, the student may explain why different initialization or training-loop control produced different convergence behavior.
Do not award if: Both implementations are present but are discussed independently, or the report merely states that one had a higher score without examining why.

- **(1 pt) Runtime and tuning.**
Award if: Computational cost is reported and the final NN configuration is supported by tuning evidence.
Do not award if: Runtime/training cost is absent, or architecture/learning-rate choices appear arbitrary.

Discussion & Conclusion (14 points)

This section should synthesize the full experiment rather than repeat model-by-model results.

- **(3 pts) Cross-model synthesis.**
Award if: At least three required algorithm families are directly compared using evidence from final metrics, curves, runtime, or error patterns, with connections to relevant supervised-learning concepts. For example, the student may explain why scaling-sensitive models behaved differently from the tree or why a flexible RBF boundary improved minority-class performance at a runtime cost.
Do not award if: Models are discussed only in isolation, the section merely ranks final scores, or claims such as "SVM performed best because SVM is powerful" are given without evidence.

- **(2 pts) Performance and efficiency trade-offs.**
Award if: At least two concrete predictive-performance versus computational-cost trade-offs are explained. Examples include Macro-F1 versus SVM fit time, kNN prediction time versus fitting simplicity, or NN stability versus training cost.
Do not award if: Runtime and predictive metrics are listed in separate tables but never compared, or the final recommendation considers predictive score alone.

- **(2 pts) Class-level behavior.**
Award if: The discussion identifies important differences among the five General Health classes, addresses less-represented classes, and explains at least one meaningful confusion pattern using precision, recall, F1, or the confusion matrix.
Do not award if: The student only discusses aggregate Accuracy, identifies a "best model" without examining minority-class behavior, or includes a confusion matrix without interpreting specific errors.

- **(2 pts) Curve-based interpretation.**
Award if: Learning and model-complexity curves support at least two specific claims about underfitting, overfitting, bias, variance, saturation, optimization, or hyperparameter sensitivity.
Do not award if: Curves are merely described visually, such as "the validation line increases," or the report makes bias/variance claims without pointing to curve evidence.

- **(2 pts) Limitations and improvements.**
Award if: At least one concrete limitation and one evidence-based follow-up experiment or improvement are identified. For example, poor minority recall may motivate class weighting, an unstable NN epoch curve may motivate a learning-rate study, or SVM sensitivity to C may motivate a broader search.
Do not award if: The student gives only generic statements such as "use more data," "try more models," or "tune better" without connecting the recommendation to observed evidence.

- **(3 pts) Conclusion synthesis.**
Award if: The conclusion concisely states what was learned, explicitly revisits the original hypothesis, addresses how class imbalance affected interpretation, discusses relevant Type I/II-style errors in the multi-class setting through observed class confusions, and identifies one realistic next step. The conclusion should emphasize what the observed performance implies rather than treating lower-performing models as failed experiments.

Do not award if: The conclusion is only a numerical recap; never revisits the hypothesis; ignores class imbalance or observed error patterns; labels models "good" or "bad" based primarily on their score; introduces results not shown earlier; or gives a generic future-work statement unrelated to the evidence.

Extra Credit: Activation Function Study (+5 points)

Extra credit must use SGD only.

Students should compare four activation functions, such as ReLU, GELU, SiLU/Swish, and tanh, using comparable architecture, preprocessing, regularization, and training procedures.

- **(2 pts) Activation curves.**
  Award if: Epoch-wise validation curves are shown for all four activation functions under comparable training conditions.
  Do not award if: One or more activation curves are missing, or architecture/training settings vary so much that the activation comparison cannot be interpreted.
- **(2 pts) Comparison table.**
  Award if: Final predictive metrics and relevant runtime/convergence information are compared across the four activation functions.
  Do not award if: The table is absent, missing major metrics, or does not allow a meaningful comparison.
- **(1 pt) Interpretation.**
  Award if: Convergence speed, stability, and performance trade-offs under SGD are discussed.
  Do not award if: The student simply identifies the highest-scoring activation without interpreting convergence or optimization behavior.

TA Reminders

- Do not grade toward an expected model winner.
- Do not grade toward an expected performance threshold. A model in the 40–60% range may be completely reasonable when the result is reproducible and well analyzed.
- Lower predictive performance often creates an opportunity for stronger analysis. Reward students who explain **why** the model struggles and what the result implies.
- Accuracy alone is insufficient for this imbalanced multiclass task.
- Compare model performance to appropriate baselines, other metrics, class-level behavior, curves, runtime, and the original hypothesis.
- Do not reward tuning on the held-out test set.
- Check carefully for target or preprocessing leakage.
- Use the examples in the rubric to calibrate decisions, but grade the underlying requirement rather than exact wording or implementation.
- The main report must contain the substantive technical argument.
- Do not treat writing style as evidence of AI use. Internal inconsistencies between prose, experiments, code, figures, citations, and provenance are substantially more meaningful.
- A high Turnitin score, AI indicator, agent artifact, or large paste is not by itself proof of misconduct.
- If provenance concerns arise, document evidence and escalate rather than assigning an ad hoc penalty.
- Submission-access problems receive normal grading feedback and a temporary zero until corrected and verified.

Version Control

- v1.0 – 09/28/2026 – TJL refactored SL rubric for Fall 2026.

Rubric written by Theodore LaGrow.